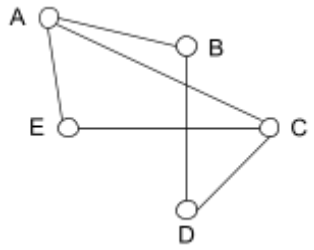
American Computer Science League

2025-2026 ● Contest 4: Short Problems ● Elementary Division

1. Graph Theory

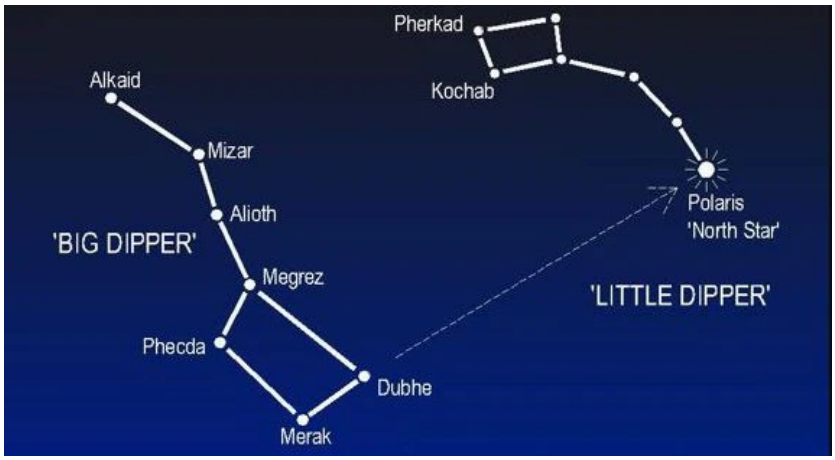
One application of graph theory is coloring graphs so that each vertex is assigned a color. If no two vertices that share an edge are assigned the same color, what is the minimum number of different colors that is required for the graph below?



A. 1
B. 2
C. 3
D. 4
E. 5

2. Graph Theory

The following image shows two constellations that are always seen in the night sky. We can think of it as a graph. Suppose one additional edge is added that connects the star Dubhe with Polaris as shown by the arrow below. This becomes a connected graph that could be traversable. Which existing edge must be removed for that new graph to be traversable?



A. Dubhe to Megrez
B. Megrez to Pheoda
C. Merak to Dubhe
D. Kochab to Pharkad
E. Merak to Pheoda

3. Graph Theory

With the advent of AI, it has become more commonplace for students to submit very similar coding solutions. The plagiarism detection algorithm discovered the following pairs of students submitted similar code:

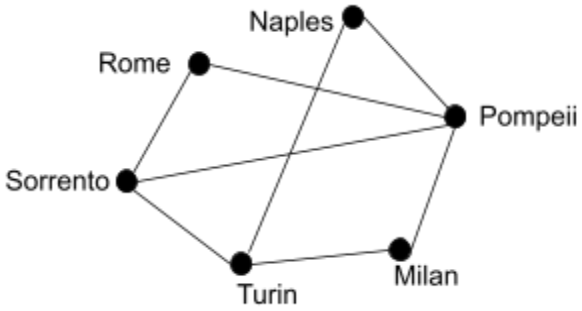
Alex and Chris, Barb and Chris, Barb and Dave, Fran and Chris, Dave and Alex, Chris and Erin, Barb and Alex, Erin and Fran.

Draw a graph that includes each of these edges. A "ring" is formed when more than 2 students form a cycle. How many different "rings" of collaborators have exactly 3 students**?**

A. 0
B. 1
C. 2
D. 3
E. 4

4. Graph Theory

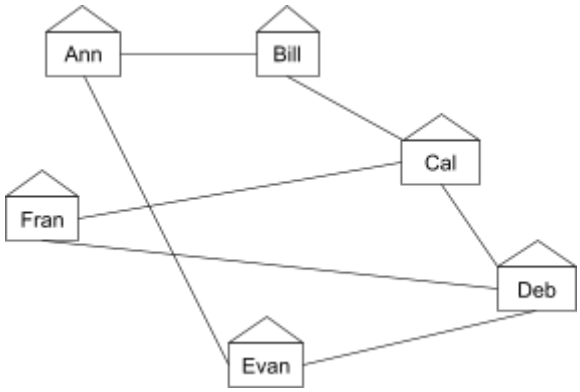
Given the graph below that represents a railroad network among 6 Italian cities (Milan, Naples, Pompeii, Rome, Sorrento, and Turin), how many ways can a person get from Milan to Rome without visiting a city twice?



A. 3
B. 4
C. 5
D. 6
E. 7

5. Graph Theory

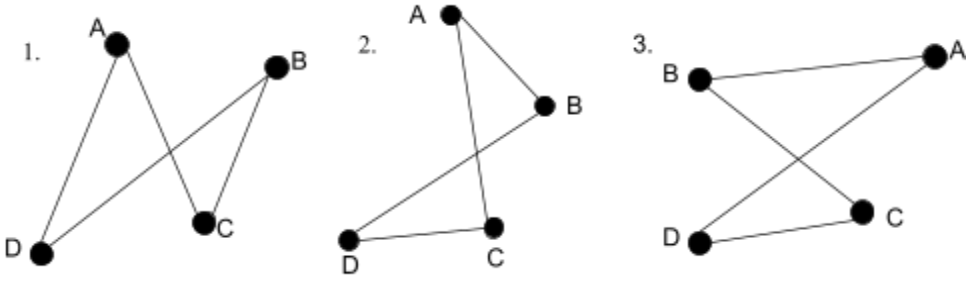
Any mapping app on your phone finds the shortest route between a starting point and a destination using the roads that exist between them. In the following graph, how many miles is the shortest route between Ann's house and Fran's house assuming that each edge is 5 miles long?



A. 5 miles
B. 10 miles
C. 15 miles
D. 20 miles
E. 25 miles

6. Graph Theory

Which of the following graphs have the same set of edges?



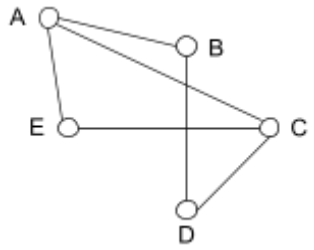
A. 1, 2
B. 2, 3
C. 1, 3
D. 1, 2, 3
E. None of them

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1. 图论

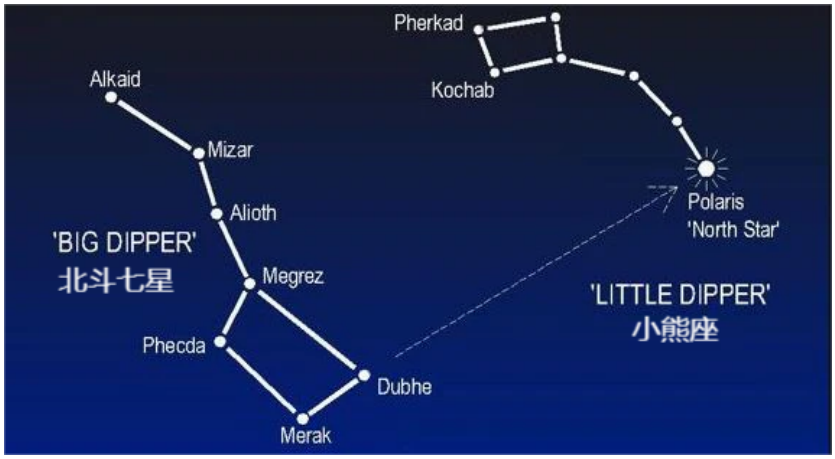
图论的一个应用是对图进行着色，即为每个顶点分配一种颜色。如果图中任意共享一条边的两个顶点颜色都不相同，那么下图最少需要多少种不同的颜色？



A. 1
B. 2
C. 3
D. 4
E. 5

2. 图论

下图展示了夜空中始终可见的两个星座。我们可以将其视为一张图。假设增加一条边，如箭头所示，连接恒星 Dubhe 与 Polaris，这样便构成一个可遍历的连通图。为了使得新图可遍历，必须移除哪一条现有边？



A. 从 Dubhe 到 Megrez
B. 从 Megrez 到 Pheoda
C. 从 Merak 到 Dubhe
D. 从 Kochab 到 Pharkad
E. 从 Merak 到 Pheoda

3. 图论

随着人工智能的出现，学生提交高度相似的编程作业变得越来越普遍。某抄袭检测算法发现以下几组学生提交了相似代码：

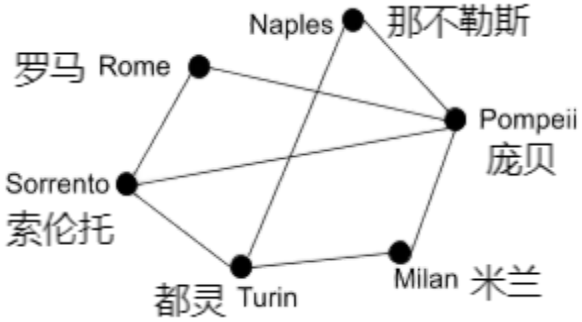
Alex 与 Chris, Barb 与 Chris, Barb 与 Dave, Fran 与 Chris, Dave 与 Alex, Chris 与 Erin, Barb 与 Alex, Erin 与 Fran.

请画出包含上述所有边的图。当超过 2 名学生构成一个环（cycle）时，即形成一个“圈”（ring）。请问有多少个正好由 3 名学生组成的圈？

A. 0
B. 1
C. 2
D. 3
E. 4

4. 图论

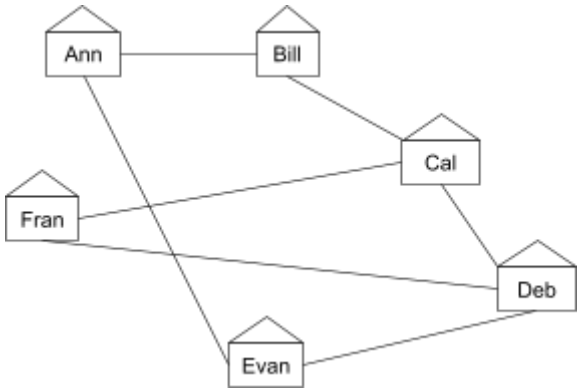
下图表示意大利六座城市（米兰、那不勒斯、庞贝、罗马、索伦托、都灵）之间的铁路网络。从米兰到罗马，不重复经过同一座城市，有多少种不同的路线？



A. 3
B. 4
C. 5
D. 6
E. 7

5. 图论

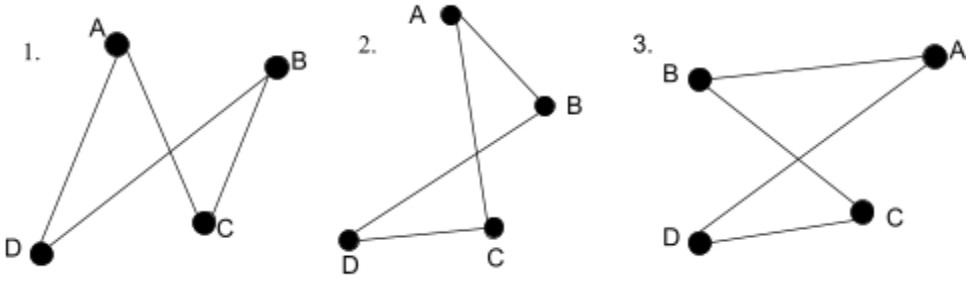
手机上的任意一款地图软件都会利用现有道路，计算起点到终点的最短路径。在下图中，假设每条边长度均为 5 英里，那么 Ann 家与 Fran 家之间的最短路径是多少英里？



A. 5 英 里
B. 10 英 里
C. 15 英 里
D. 20 英 里
E. 25 英 里

6. 图论

下列哪些图具有相同的边集？



A. 1, 2
B. 2, 3
C. 1, 3
D. 1, 2, 3
E. 以上**均不是**